RAG

1. What is RAG?
2. Problems RAG solves - What? Why we need RAG
3. Explain RAG working end to end in step by step.
4. What are 2 major phases we have for Rag. Explain them and the component's it has.
5. In a RAG pipeline what are the different stages.
6. Explain Loaders - What are they, what are different type of document sources. Why loaders are needed.
7. What are the most common open source available tools available to load pdf.
8. Is there an open source tool that extracts the document hierarchy/structure like headers, section, paragraph. Also tells if there is a tables or image in a given page.
9. In case tables are in the pdf - how to handle them. What if the tables span multiple pages, how will we handle it then.
10. How to handle images in a pdf.
11. What is chunking and why it is needed.
12. What is trade off of doing chunking
13. Explain different chunking strategies. Go from basic to advance. Have examples and proper deep explanation.
14. What are the best practices to follow for chunking.
15. What is embedding and why it is needed.
16. What are the different types of embeddings - Sparse, dense.
17. Explain ~~to~~ sparse embeddings. What are different techniques to this. And What is the most common one.

18. Explain dense Embedding, How are they are created.
19. ~~st~~ Explain hybrid embedding with examples.
20. Explain ~~Embed~~ Indexing properly.
21. Compare Indexing in vector db to table in SQL db.
22. What all information is provided to create an index
23. Why we need vector db. (Persistance, size etc)
24. What are the different type of Index. Explanation with examples.
25. Which index type is most commonly used. And it explained in detail.
26. HNSW explained, with an simple example. ~~How we~~ How will the ~~in~~ data structure will be created ~~ahead~~ in the 'offline' stage with a simple example step by step.
27. What are the different similarity metrics. Compare when to use what.
28. Explain Vector DB and what is optimised in it
29. What are the most ~~for~~ famous vector databases and what are the differences between them. When to use which.
30. What are the open source Vector DB that is good and can be deployed in an on prem docker.
31. How search happens in Vector DB.
32. Explain Hybrid search and How it combines dense and sparse search.
33. What is RRF and How it works with an example.
34. Explain metadata and how it is used for filtering
35. What is the retrieval process.
36. Explain different retrieval strategies. be thorough and explain in depth.

37. Explain Re Ranker and Why they are needed.
38. Where does the re ranker sit in the pipeline. (Architecture)
39. Different types of re ranker explained
40. The most popular re ranker explained in depth with example.
41. Prompting for Rag.
42. Different prompting techniques used in RAG.
43. How to add citation. What is citation and why it is needed.
44. Explain advance prompting for RAG - Query Reformation, decomposition, history, output formatting, HyDE etc. (Knowledge Graph)
45. Explain generation process
46. Different models and there parameters
47. How to control Hallucination
48. How to optimise cost and token consumption
49. How to evaluate RAG.
50. What are the different metric to evaluate it
51. All the metrics explained with examples and How we can calculate them.
52. Explain the reference free and reference based evaluation. What is the difference. What are metrics within it and How to calculate them